

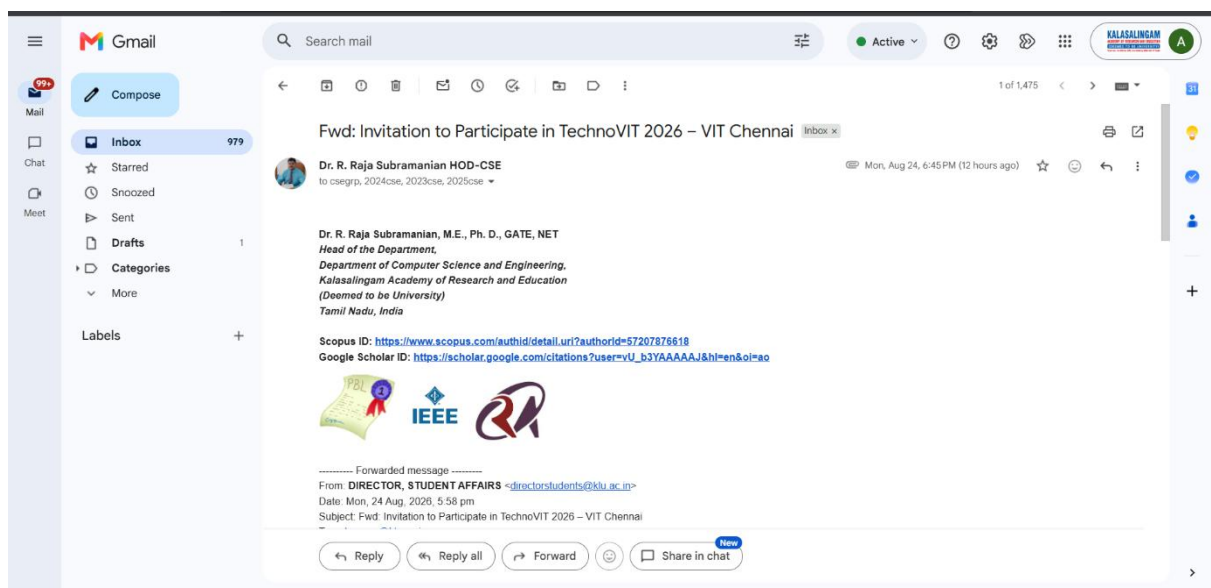
Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."



2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

Original Message

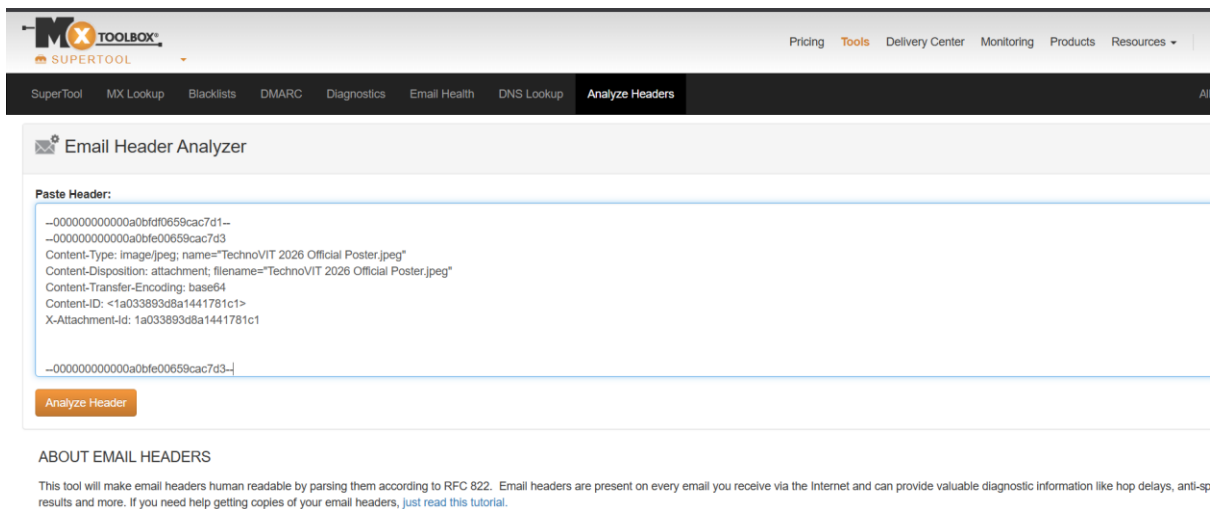
Message ID	<CAMq6mVazV=hG+d17aJQDG_h-YLj77xtAIT69e7-D+LR2cwZCpg@mail.gmail.com>
Created at:	Mon, Aug 24, 2026 at 6:45 PM (Delivered after 27 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
Subject:	Fwd: Invitation to Participate in TechnoVIT 2026 – VIT Chennai
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain klu.ac.in Learn more
DMARC:	'PASS' Learn more

[Download Original](#)[Copy to clipboard](#)

Delivered-To: 99240041414@klu.ac.in
Received: by 2002:a05:7301:30c:b0:327:c17f:fa3b with SMTP id ia12csp370607dyb; Mon, 24 Aug 2026 06:15:35 -0700 (PDT)
X-Forwarded-Encrypted: i=5; AHghRpxaVc2saafYVvubtFbuc6gaiv/jbF6Hho7eCaccqH7377AqIsEzYjo66zNT18DLIP8gBabX90wZ6rwQ==@klu.ac.in
X-Received: by 2002:a05:6402:505b:b0:69c:811f:8f5 with SMTP id 4fb4d7f45d1cf-6a42f1b9325mr30805415a12.6.1787577335463; Mon, 24 Aug 2026 06:15:35 -0700 (PDT)
ARC-Source: i=4; a=rsa-sha256; t=1787577335; cv=pass; d=google.com; s=arc-20260327; b=1t0e68HedvZaowM4VpqbVzt27Cmbf9vcr3hVv1ohgpmgQZmHAqzXlplCmaue18 mFDeY610L124MVA82oabXay9a+HkL1gBELHMKO3htf16kegi+gEz3+vBrbc+ai+ICtc 5fG5sqsaP3oz711b/s7KQ508m2MJo7uoc3C2x1IR310C4F/t1p2bYnLRV51VT7vGuW 5lYksV/MEpJ12EpspnZ2013Y6uJdp6DIgokR+3u/0Twa3yQKvY2rptv1+0VLtGzU14eK UB5H5ZTC1X0F6g/2K1Z0Hvazm8tN6LJRAHkN/I4p50h6112dFjmg9qbro3u1XJHY 6SYQ==
ARC-Message-Signature: i=4; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe:list-archive:list-help:list-post:list-id

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.



The screenshot shows the MX Toolbox website's 'Email Header Analyzer' tool. The header includes the MX Toolbox logo and navigation links like 'Pricing', 'Tools', 'Delivery Center', 'Monitoring', 'Products', and 'Resources'. Below the navigation bar, there's a menu with 'SuperTool', 'MX Lookup', 'Blacklists', 'DMARC', 'Diagnostics', 'Email Health', 'DNS Lookup', and 'Analyze Headers'. The main section is titled 'Email Header Analyzer' and contains a text area for pasting email headers. The pasted header text is as follows:

```
--00000000000a0bfe00659cac7d1--
--00000000000a0bfe00659cac7d3
Content-Type: image/jpeg; name="TechnoVIT 2026 Official Poster.jpeg"
Content-Disposition: attachment; filename="TechnoVIT 2026 Official Poster.jpeg"
Content-Transfer-Encoding: base64
Content-ID: <1a033893d8a1441781c1>
X-Attachment-Id: 1a033893d8a1441781c1
--00000000000a0bfe00659cac7d3--
```

Below the text area is an 'Analyze Header' button. At the bottom, there's a section titled 'ABOUT EMAIL HEADERS' with a paragraph explaining the tool's purpose: 'This tool will make email headers human readable by parsing them according to RFC 822. Email headers are present on every email you receive via the Internet and can provide valuable diagnostic information like hop delays, anti-spam results and more. If you need help getting copies of your email headers, [just read this tutorial](#).'

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**



5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	mail-sor-f41.google.com 209.85.220.41	mx.google.com	SMTPS	8/24/2026 1:15:28 PM	✓
2	1 Second		2002:a05:6402:3130:b0:697:59b1:5eb9	SMTP	8/24/2026 1:15:29 PM	
3	6 seconds	mail-sor-f69.google.com 209.85.220.69	mx.google.com	SMTPS	8/24/2026 1:15:35 PM	✓
4	0 seconds		2002:a05:7301:30c:b0:327:c171:fa3b	SMTP	8/24/2026 1:15:35 PM	

EmailHeaders advertisement image

SPF and DKIM Information

dmARC:klu.ac.in [Show](#) [Solve Email Delivery Problems](#)

v=DMARC1;p=quarantine;sp=none;adkim=r;aspf=r;pct=100;fo=0;rfa=afrr;ri=86400

spf:klu.ac.in:209.85.220.69 [Show](#) [Solve Email Delivery Problems](#)

vsfpf1 ip4:208.115.225.178 include:antispamcloud.com +include:_spf.google.com ~all

dkim:klu.ac.in:google [Show](#)

6. Examine the SPF, DKIM, and DMARC Results

- SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - Pass:** Indicates the IP is authorized.
 - Fail:** The IP is not authorized and could be a sign of spoofing.
- DKIM:** Verifies that the email content has not been altered in transit.
 - Pass:** The email has not been tampered with.
 - Fail:** The email content may have been altered.
- DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - Pass:** Indicates alignment with SPF/DKIM policies.
 - Fail:** Indicates possible spoofing or tampering.

7. Analyze the ‘Message-ID’

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

```
dkim:klu.ac.in:google Show

DKIM Public Record:
v=DKIM1; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQFAAQACIIB8ICGKCAQEAqsF7ayERWQk3tAd6K15oq14ud/BKVZY1yueFTXSnhQ12E0L1rPwQcYyHsn7onBpano9Zc5Dr7x1XC5Ed3X16p5a44TcnMP/SgLVaGdA1lP84qh45Hb8HE5mlap

DKIM Signature:
v=1; a=rsa-sha256; c=relaxed/relaxed; d=klu.ac.in; s=google; t=1787577335; x=1788182135; darn=klu.ac.in; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-list:precedence-x-
```

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.